

Transforming Agriculture with AI: Automated Crop Insect Identification Using InceptionV3 Transfer Learning

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Abstract— Identifying crop insects and classifying them is a challenging task for agriculturalists. A transfer learning method is suggested in this paper to address this issue. The proposed approach uses InceptionV3 as a base deep learning model and it is given additional custom layers. The model is fine-tuned using a custom dataset made up of five classes with 1000 insects each. The insect and pest images available from the National Bureau of Agricultural Insect Resources datasets and the Xie1 dataset are used for testing the model. Preprocessing techniques are applied to resize the images, and data augmentation is employed to prevent overfitting. The model was analyzed on the validation dataset with Adam, Adagrad, and Nadam optimizers, and obtained recognition accuracies of 86.6%, 77.8%, and 88.56% respectively. The outcomes confirms that the model performs well with Nadam optimizer for insect identification.

Keywords— InceptionV3, Deep Learning, Transfer Learning, Insect Detection, Smart Agriculture

I. INTRODUCTION

Due to the undistinguishable appearance in the initial stages of growth, precisely classifying and identifying all varieties of agricultural insects is a difficult task for the formers. Identifying pests and insects provides valuable data for pest management planning and it is a part of intelligent agriculture assistive system. An intelligent agricultural assistive system refers to the integration of advanced technologies and data analysis techniques to enhance and optimize agricultural practices with low cost and high efficiency. These systems use a range of technologies, including Internet of Things, sensors, data analytics, drones, and artificial intelligence (AI) techniques to automate farming operations and provide real-time monitoring and decision-making. Since different pests and insects require different control methods, by accurately identifying the pest or insect species present, appropriate control strategies can be developed and applied [1]. Different

pests and insects cause varying levels of damage to crops, structures, and property. Identifying the specific pests or insects responsible for fast and high levels of damage at early stages leads to timely actions to mitigate the damage and protect valuable resources. Certain pests and insects can transmit diseases to humans, animals, and plants. Identifying the presence of disease-carrying insects allows for appropriate preventive measures, such as insecticide application or habitat modification, to reduce the risk of disease transmission. Some insects and pests can have significant ecological impacts. For instance, invasive species can disrupt ecosystems by outcompeting native species or damaging habitats. In summary, identifying pests and insects is crucial for effective pest control, damage prevention, disease management, environmental conservation, and sustainable pest management practices.

Image analytic-based pest monitoring algorithms are affected due to the fluctuation in appearance, clustering backdrop and characteristics of images. It is exceedingly challenging to manually choose the features for pests of small sizes and hazy features. Because of advancements in deep learning, it is now probable to automatically identify pests and insects from subpar images. The key goal of the research work carried out in this paper is to develop an efficient deep-learning system with a suitable optimizer for classifying insects.

The application of AI and transfer learning in automated crop insect identification can have several benefits. It can expedite the identification process, allowing for quicker responses to pest infestations. It can also reduce reliance on human expertise and potentially enable non-experts to identify pests accurately. Additionally, the data collected during the identification process can be used for monitoring and analyzing pest patterns, leading to better pest management strategies.

The rest of the manuscript is arranged as follows. An overview of the literature regarding pest and insect detection is

given in section 2. The suggested method and the benchmark datasets employed are presented in section 3; the investigational findings are discussed in section 4; and lastly, the paper closes with conclusion.

II. LITERATURE SURVEY

Insect and pest detection is essential in food security and overall quality of life. It is also very important for a steady agricultural economy. Real-world field circumstances, complex environments, and various vantage points and poses present significant obstacles to identifying and detecting pests in crops. The voting ensemble technique suggested by Zeba Anwara and Sarfaraz Masood [1] using VGG16, VGG19 and ResNet 50 and yielded 82.5% recognition accuracy. The SIFT-HMAX model developed by Limiao Deng et al. [2] provided an accuracy of 85.5%. This model improves the Hierarchical Model by combining Scale Invariant Feature Transform (SIFT) and it was named as SIFT-HMAX. Even though HMAX was providing outstanding performance in object recognition, it was sensitive to rotation, but the integration of SIFT made it rotational invariant. The authors generated saliency maps by applying Saliency Using Natural Statistics Model (SUN) and detected regions of interest in the pest images. The fusion of features extracted from models like ResNetV2 and InceptionV3 has been explored by Varshney et al. [3] for classifying Madhubani Arts. These studies demonstrate the ongoing research and development efforts focusing on transfer learning systems.

Wenyong Li et al. [4] constructed a transfer learning model based on VGG16 pre-trained on the COCO dataset [5] and named it TPest-RCNN to detect whiteflies and trips from images captured from a yellow sticky trap. On a validation set, this model attained mean precision and mean F1 scores of 0.944 and 0.952, respectively. The model was reliable for finding microscopic pests on pictures with various pest concentrations and light reflections. A linear regression was made between their model's automatic detection and the manual counts for 41 days, and the resulting determination coefficients for whiteflies and thrips, were 99.6% and 97.4% respectively. Yan Li et al. [6] suggested using multifractal analysis to find tiny pests. This technique, which relies on both local image singularity and global visual attributes, was used to identify Whitefly. They obtained 88.6% of the F1 measure. A CNN-based pest monitoring system was suggested by Fangyuan Wang et al. [7] to find and count rice planthoppers. They employed a sensitive score matrix (SSN) to improve the bounding box regression and classification results and achieved 91% recognition recall. They employed three different scales (32px, 64px, 128px) on each pixel with three aspect ratios (1x1, 1x2, 2x1) such that nine rice planthopper regions are predicted. They implemented their model on Caffe and used RMSprop as an optimizer and employed early stopping to choose the optimum iteration for training. To avoid overfitting they applied a 0.5 dropout rate. They fixed the learning rate to 0.001 and reduced it by 10% for every 50,000 iterations. They set the minibatch size to 1.

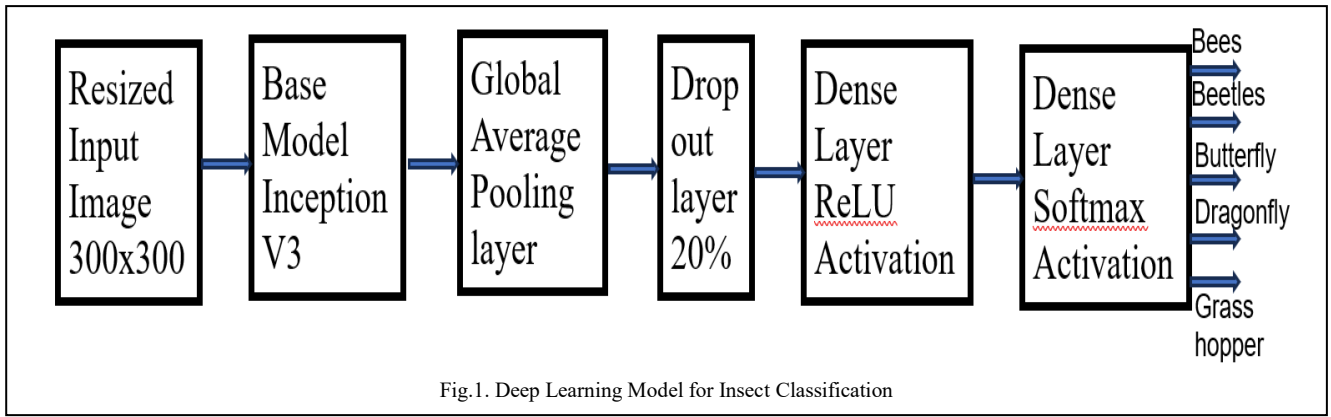
They used ZF, ResNet-50, VGG-16, and ResNet 101 as backbone models. They adopted the approximate joint strategy to train the SSM module along with the CNN. Muppala and Guruviah [8] proposed a neural network with four layers and identified leaf folders and yellow stem borers by applying a search and rescue optimization. They enhanced the images captured by light traps using the deer hunting algorithm, segmented them by using social ski-driver optimization and removed the impulse noise using a fast average group filter.

The performance of a deep learning model decreases when the layers are increased. This problem is solved in the ResNet by the skip connection that will transmit the signal from any input layer to a higher layer. But it does not have cross-channel interaction. This is solved by J. Hu [9] in Squeeze and Excitation Network (SE-Net) by changing the residual block using a cross-channel attention mechanism. This enables the network focus on local features. Grouping the channels will let to focus each group of channels on different features and this idea is used in ResNext by Saining Xie [10]. The SENet and the ResNet were combined by Hang Zhang et al. in [11] and they got better segmentation and classification performance. Xinting Yang et al. [12] suggested a CNN model with spatial and channel attention mechanisms and recognized the locations of pests accurately in the field images.

Thenmozhi and Srinivasulu Reddy [13] applied the concept of sharing weights in Deep CNN to reduce the vanishing gradient problem. By utilizing image pre-processing techniques such as edge finding and noise reduction, they improved the task of classifying insects. An edge-detected binary insect image was determined to have the most outermost contours. The top-left coordinate (x,y) and the width and height (w,h) serve as the four points that make up the bounding box and it is retrieved from the RGB image using the coordinates (x+w, y+h). After insect images are cropped they are resized to 227×227 pixels. Their model successfully classified the 40 classes of insects provided in the NBAIR insect dataset [14], 24 classes from the Xie1 insect dataset [15], and 40 classes provided in the Xie2 insect dataset [16] with the highest classification accuracy.

III. MATERIALS AND METHODS

Farmers find it challenging to correctly classify and identify every variety of crop bugs, so this research offers a strategy based on transfer learning to automate this process. InceptionV3, a pre-trained model on the ImageNet dataset is the base model used in this research. The ImageNet dataset consists of millions of labeled images across thousands of different classes. Transfer learning is applied by adding custom layers to this InceptionV3 and the model is fine-tuned to identify pests as shown in Figure 1. Five classes with 1000 insects in each class were collected from the benchmark dataset IP102 [17] to fine-tune the InceptionV3 model. The ICAR-National Bureau of Agricultural Insect Resources (NBAIR) datasets [14], Xie1 [16] were used to test the model



The feature extraction occurs up to the bottleneck layer, which is the last layer before the flattening layer in the InceptionV3 model. Preprocessing is done to resize the image such that it is suitable for giving to the input layer of the base model. Since a smaller number of images i.e., 5000 images are used for training, data augmentation techniques with scaling, rotation, translation and data reflection were used to avoid the network from overfitting. The convolution layers of the base model extract features from the images. Down sampling is done via the pooling layers and they keep the most crucial details of the insect images. This layer inhibits overfitting by reducing the number of parameters and the representation's spatial size [8]. The last or top layer i.e. the head of the base model is a fully-connected layer but this layer is removed in the proposed model. A dense layer with ReLU activation is created and the Global Average Pooling layer's output is fed to this dense layer with a dropout of 20%. Another dense layer with an activation function SoftMax is added to the network. This layer acts as a prediction layer and it provides five outputs. These outputs represent the probabilities of the five classes predicted for the given input image. The class that has the highest probability is assigned to that the image.

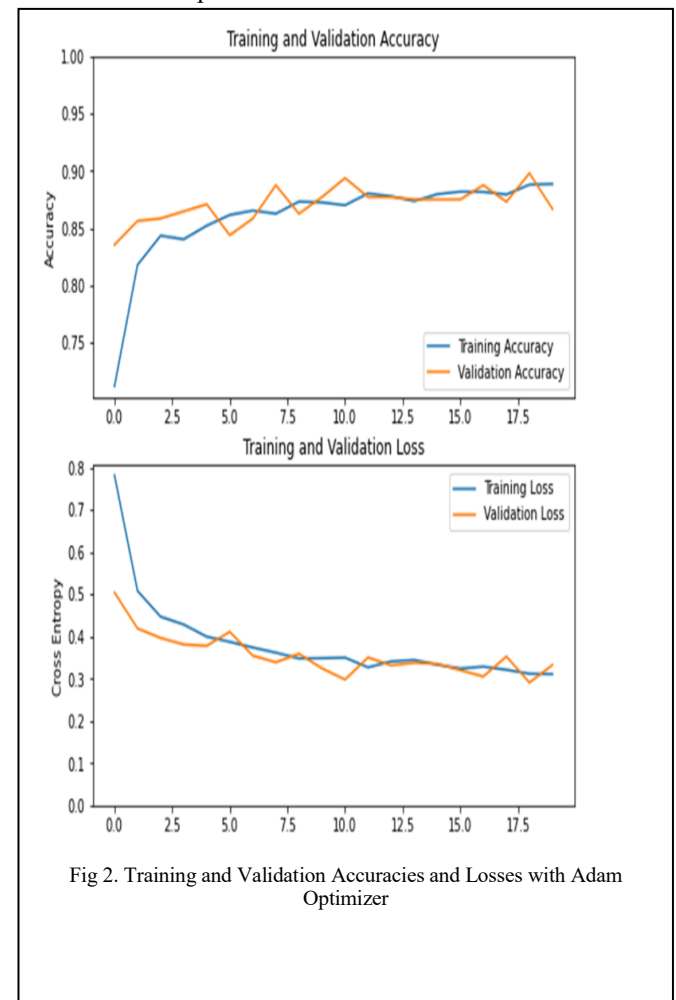
Different optimizers with a same learning rate 0.0001 was applied to train this transfer learning model and the better optimizer was chosen. Binary cross entropy was used to estimate the loss. The images are divided into batches with each batch containing 32 images training and validation. Out of 5000 images, 90% of images are used to train and 500 remaining images are used for validation. The weights of the base model are frozen. One batch from the training dataset is taken along with its labels and passed through the base model. This creates features of size (8,8,2048). Then a Global Average Pooling is applied on this feature map to reduce the dimension to 2048.

IV. EXPERIMENT AND PERFORMANCE ANALYSIS

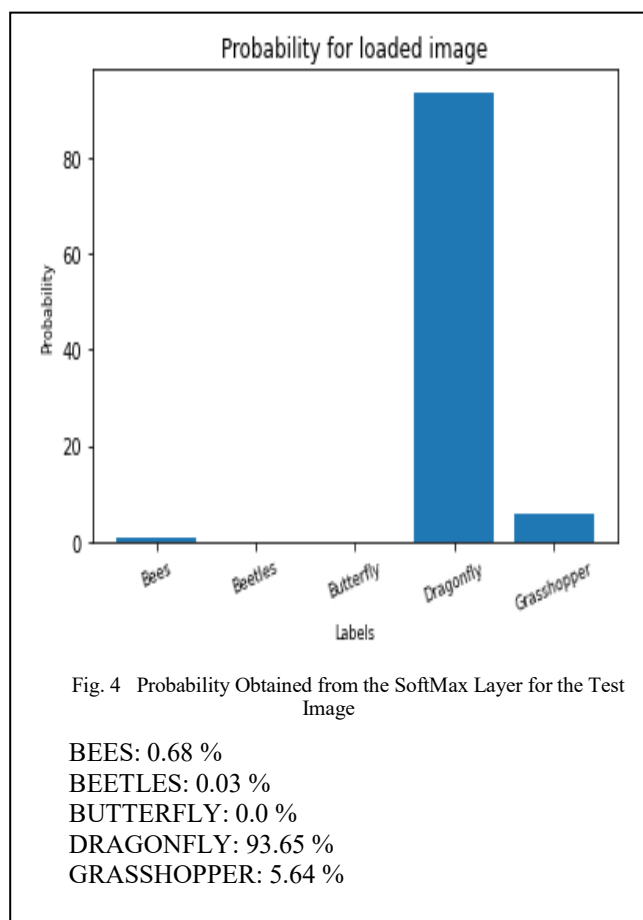
A. Experiment 1: Adam Optimization

In this experiment, the Adam optimizer with 0.0001 learning rate was used for training the transfer learning model and it was trained on the insect dataset discussed in the

previous section using an HP Pavilion system that has a core i7 10th-generation CPU of 2.66 GHz and a memory of 32GB. The batch size was fixed to 32 and 20 epochs were used. Since 4500 images were used for training, the steps used per epoch is 140 (4500//32=140). The number of trainable parameters is 21,802,784. Even though the number of epochs was set to 20, early stopping was used and it was found that the accuracy was maximum 18th epoch and its value was 86.6% and the loss at

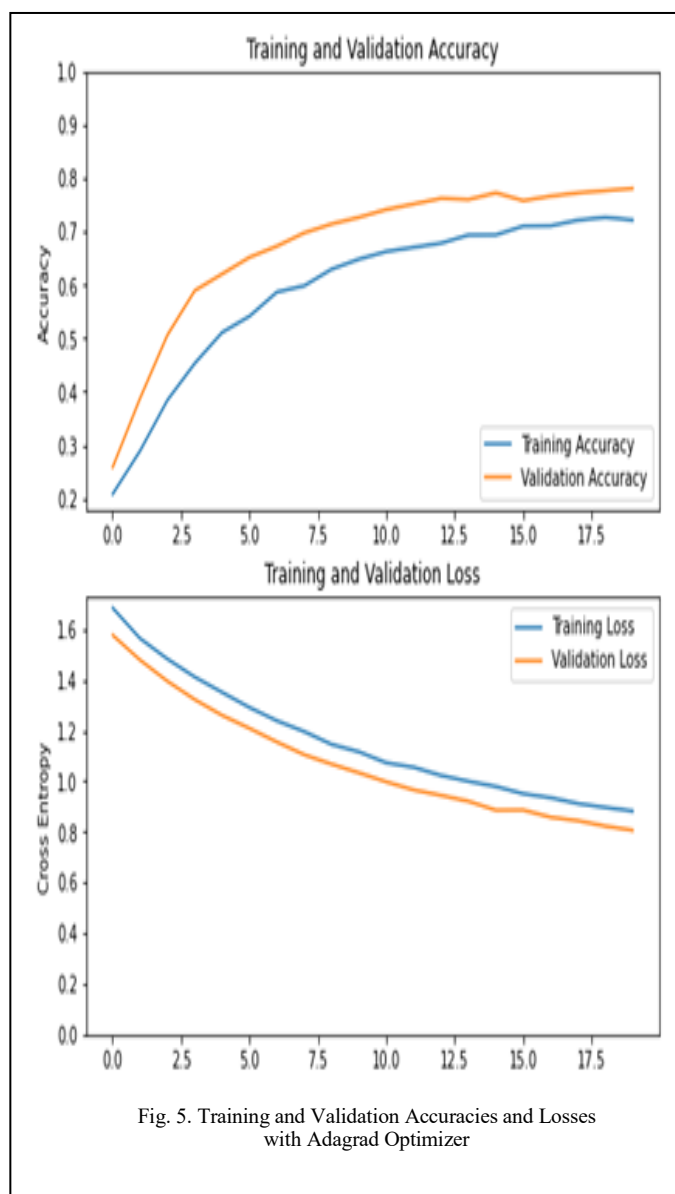


this accuracy was 0.3362. The test accuracy obtained was 86.599% (figure 2). The recognition probabilities obtained for the test image (Figure 3) are given in Figure 4.



Experiment 2: Adagrad Optimization

In this experiment, Adagrad optimization was used without changing other parameters of the model. The same dataset was given as input and the performance was observed. Fig. 5 shows that the accuracy is increasing and the losses are decreasing smoothly during training and validation. But in the case of the Adam optimization, the values are oscillating. It was also observed that at any epoch Adagrad has provided less accuracy compared to Adam optimization. In this experiment, Adagrad has provided a recognition accuracy of 77.8% and a loss of 0.8103 at the 18th iteration whereas Adam optimization has provided 86.6% accuracy and a loss of 0.3362 at the same iteration. When the dragonfly shown in Figure 2 was fed as input with Adagrad optimization, the probabilities obtained for recognizing the insect are shown in Figure 6. Even though the network recognizes the insect, it does not differentiate it from other insect classes much. So, Adam optimization is preferred to use in this model.



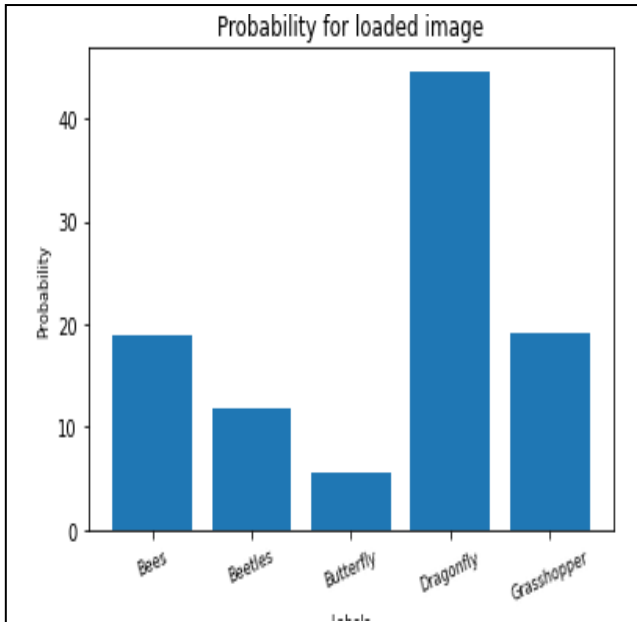


Fig. 6 Probability Obtained from the SoftMax Layer for the Test Image (Adagrad)

BEES: 18.87 %
 BEETLES: 11.83 %
 BUTTERFLY: 5.45 %
 DRAGONFLY: 44.63 %
 GRASSHOPPER: 19.21 %



Fig. 7. Training and Validation Accuracies and Losses with Nadam Optimizer

B. Experiment 3: Nadam Optimization

In this experiment, the Nadam optimizer was employed in the proposed model. Nadam optimizer combines the benefits of Nesterov Accelerated Gradient (NAG) and Adam. NAG helps accelerate convergence by considering the current gradient along with the momentum from the previous step. Adam, on the other hand, is adaptively choosing the learning rate based on its past gradients for each parameter individually. Nadam optimizer incorporates the NAG updates into the adaptive learning rate scheme of Adam. It uses NAG to calculate the gradient at the updated value of the parameters, which helps improve convergence. Nadam aims to provide the benefits of both approaches, including fast convergence and robustness to different learning rate settings. So this optimization allows the network to quickly reach the maximum accuracy here at the 8th epoch and loss also reduces at this stage as shown in figure 7. After this point, using more epochs increases the accuracy slowly. At the 18th epoch, it shows a small improvement in accuracy (88.56%) compared to Adam optimizers. The loss at this point is 0.3202. When the trained model was used to predict the test image shown in Figure 3, it was predicted correctly with a probability of 83.92 %. Due to the fast convergence, the Nadam optimizer is preferred to train the model designed in this paper.

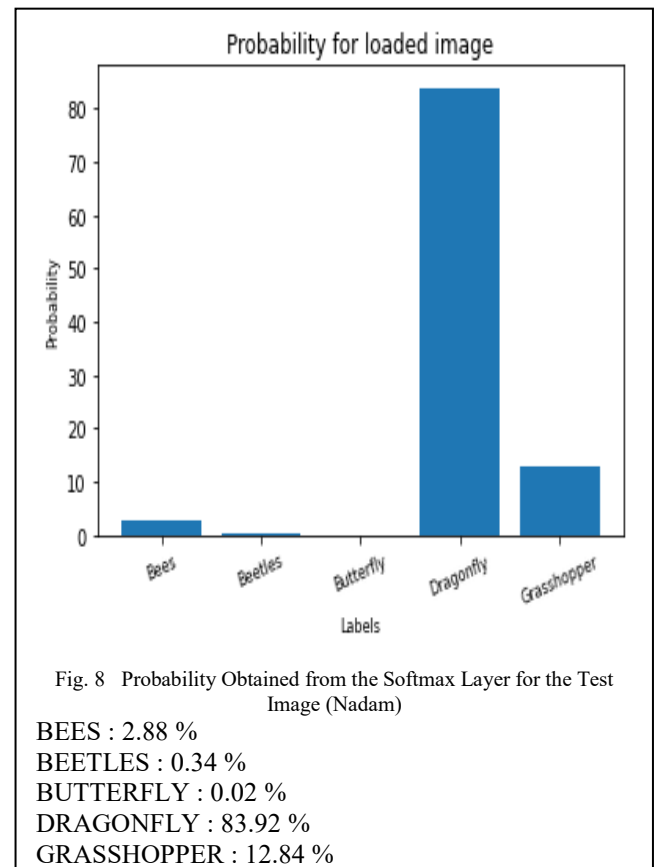


Fig. 8 Probability Obtained from the Softmax Layer for the Test Image (Nadam)

BEES : 2.88 %
 BEETLES : 0.34 %
 BUTTERFLY : 0.02 %
 DRAGONFLY : 83.92 %
 GRASSHOPPER : 12.84 %

TABLE 1: COMPARISON OF RECOGNITION ACCURACY OF THE PROPOSED MODEL

S.No	Researchers	Recognition Accuracy
1	Z. Anwar and S. Masood [1]	82.5%
2	L. Deng et al. [2]	85.5%
3	W. Li et al. [4]	94.4% (Precision – COCO Dataset)
4	Yan Li et al. [6]	88.6% (F1-measure for Whitefly recognition)
5	Proposed Method	
	Adam	86.6%
	Adagrad	77.8%
	Nadam	88.56%

The accuracy for pest recognition of the proposed model is comparable to the results provided by other researchers (Table 1). However, serial number 3 shows the highest value but they have obtained only the precision (not accuracy) using the COCO dataset. Similarly, Yan Li et al. have shown the F1 measure for whitefly recognition. In the model developed in this work, Dragonfly has been recognized with an accuracy of 93.65%.

V. CONCLUSION

The experimental results demonstrated the influence of optimizers in the proposed approach. The transfer learning model attained an accuracy of 88.56% on the test dataset with the Nadam optimization, showcasing its ability to accurately classify crop insects. The highest recognition probability of 93.65% was obtained for the class "Dragonfly," when the Adam optimizer was used and hence correctly identified the insect in the given image. The use of preprocessing techniques, such as resizing the images and data augmentation, proved beneficial in enhancing the model's performance and preventing overfitting. The application of a dropout of 20% and a SoftMax activation cause an efficient classification of the extracted features. The proposed method has significant practical implications for farmers and agriculturalists. By automating the classification and identification of crop insects, it offers valuable support to farmers, enabling them to promptly detect and address pest-related issues. The transfer learning approach presented in this paper can be further explored and extended to handle other agricultural pest identification tasks, contributing to improved crop management and increased agricultural productivity.

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